

Swarup Das

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Computer Science Undergrad seeking Software Engineering roles.

Education

Siksha 'O' Anusandhan University, ITER

Bachelor of Technology in Computer Science Engineering

Bhubaneswar, Odisha

Nov. 2023 – Present

Projects

SentinelPay | Spring Boot, Apache Kafka, Redis, SAGA, MySQL, Docker

June 2026 – Aug 2026

<https://github.com/carbon-seven/SentinelPay>

- Engineered a distributed microservices banking platform (Account, Transaction Orchestrator, Fraud Detection, Payment, Notification) routed through **Spring Cloud API Gateway** with token-bucket rate limiting.
- Implemented the **SAGA Orchestration Pattern** to guarantee eventual consistency across distributed databases, executing automated compensating refund workflows upon transaction failures.
- Built asynchronous event streaming pipelines using **Apache Kafka** to publish and consume transaction lifecycle events (*initiated, completed, refunded*) to a dedicated fraud engine.
- Designed multi-layered fraud detection algorithms using **Redis** for sliding-window velocity rate limiting, volume multiplier spike detection, and account balance drainage prevention.
- Integrated dynamic OTP verification and payment gateway webhooks with **idempotency guards** to eliminate duplicate balance deductions.

SRCTC | Java, Spring Boot, MySQL, REST APIs, Maven, React

April 2025 – May 2025

<https://github.com/carbon-seven/StrawPlate>

- Architected a modular backend replicating core IRCTC booking workflows, including seat availability checking, booking management, and cancellation logic.
- Designed normalized relational database schemas in **MySQL**, optimizing queries via **Spring Data JPA** for high-concurrency ticket reservations.
- Implemented secure token-based authentication (**JWT**) and role-based access control (**RBAC**) for passenger and administrative access.
- Developed RESTful API endpoints and integrated them seamlessly with a responsive React frontend.

Jarwizz | Python, Speech Recognition, OS Automation, Multi-threading

Jan 2026 – March 2026

<https://github.com/carbon-seven/Jarwizz>

- Engineered an automated desktop assistant capable of parsing voice commands for local task automation, web searches, and application launching.
- Integrated custom voice I/O pipelines using multi-threading to handle audio capture and synthesis without UI latency.

Technical Skills

Languages: Java, Python, C/C++, SQL, JavaScript, HTML/CSS, Bash

Frameworks & Backend: Spring Boot, Spring Cloud, Spring Data JPA, Microservices, SAGA Pattern, REST APIs, React

Messaging & Caching: Apache Kafka, Redis

Developer Tools: Git, GitHub, Docker, Maven, Gradle, Postman, IntelliJ IDEA, VS Code

Platforms: Linux (Arch, Ubuntu), Windows